# Lab 6 Password Policy Coverage Analysis

## Final Laboratory Report

This report analyzes the assigned L6 L7 39 password policy checker. The completed suites raise the starter result from 52% combined coverage to 91% statement coverage and then to 100% statement and branch coverage.

**Student Name:** _____

**Student ID:** _____

**Course:** _____

**Instructor:** _____

**Submission Date:** 1 November 2026

# Objective

The lab objective was to measure the assigned module with coverage.py, identify untested statements and decision outcomes, and write targeted pytest cases that exceed 90% code coverage and 90% branch coverage.

# Assigned Repository

**Source project:** https://github.com/SETAPESU26/L6_L7_39_password_policy

# Application Under Test

| Function | Responsibility | Important decisions |
|---|---|---|
| check_password | Returns policy violations | Length, common password, and four character classes |
| strength | Rates the password | Strong, good, weak, or very weak |
| suggest_fix | Suggests the first repair | No issue, short, common, missing class, or fallback |

# Initial Coverage Analysis

The assigned starter file contains two valid-password tests. With branch tracking enabled, both pass, but the module reaches only 52% combined coverage: 21 of 47 statements are missed and nine branches are partial. The untested behavior includes length failures, common passwords, missing character classes, alternative strength ratings, and every suggestion outcome.

| Run | Tests | Statements missed | Branch status | Coverage |
|---|---|---|---|---|
| Starter tests | 2 passed | 21 of 47 | 9 partial | 52% |
| Code suite | 6 passed | 4 of 47 | Not required | 91% line |
| Branch suite | 17 passed | 0 of 47 | 36 of 36 | 100% |

# Test Design

- Exercise valid, short, common, and missing-character-class passwords for the code-coverage milestone.
- Add the over-64-character boundary and every strength return for complete decision coverage.
- Exercise all five suggest_fix outcomes, including the empty list and fallback branch.
- Use pytest parameterization so related branch cases remain compact and readable.

# Code Coverage Analysis

The code-focused suite contains six tests covering valid passwords, the short and common rules, all four character-class checks, weak and very weak ratings, and a missing-class suggestion. It exceeds the lab threshold with 91% statement coverage.

## Commands

```
coverage erase
coverage run --source=password_policy -m pytest -q CodeCoverageTest_39_password_policy.py
coverage report -m
```

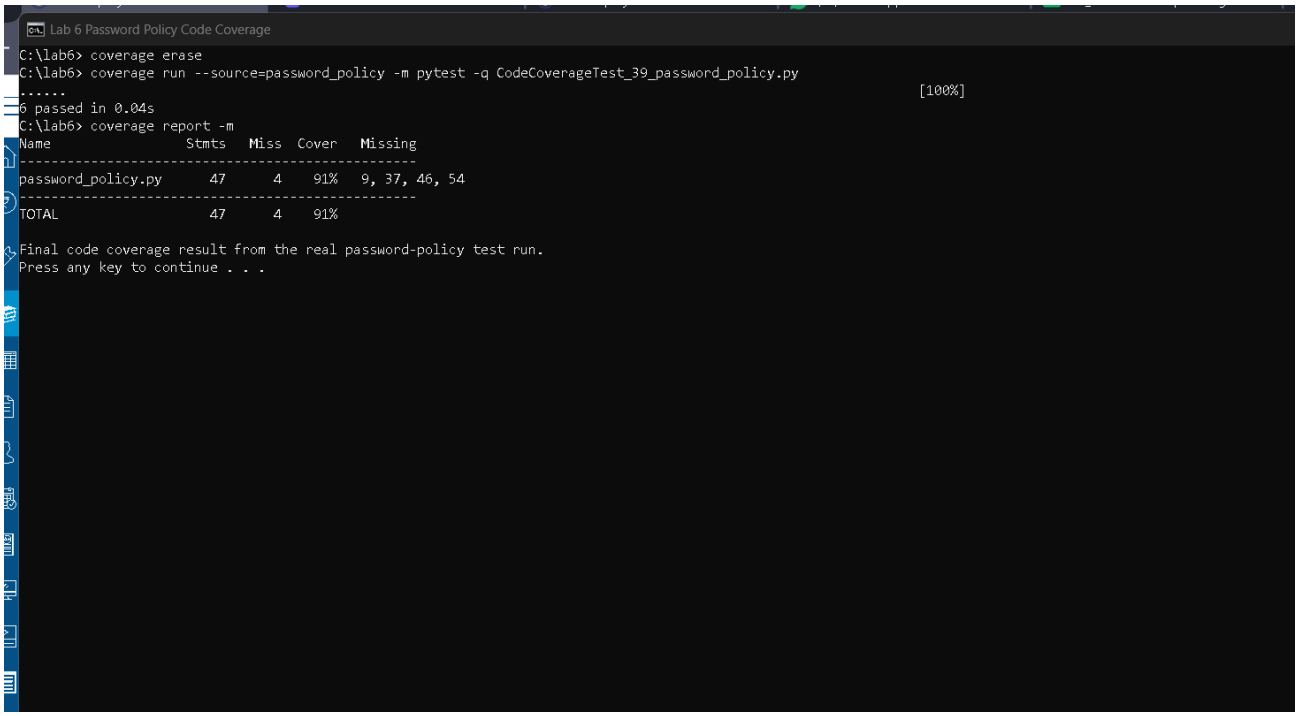



*Figure 1 Genuine terminal screenshot showing 6 passing tests and 91% code coverage*

The report identifies lines 9, 37, 46, and 54 as unexecuted. These correspond to the too-long rule, the strong rating, the no-fix-needed return, and the fallback suggestion. Those remaining outcomes guided the branch-focused additions.

# Branch Coverage Analysis

The branch suite expands the coverage set to 17 parameterized cases. It evaluates both outcomes of the length and common-password conditions, every character classification path, all four strength results, and all five suggestion paths.

| Area | Cases included | Outcome |
|---|---|---|
| Password issues | Short, long, common, valid, and each missing class | All policy decisions |
| Strength | Strong, good, weak, and very weak | All four returns |
| Suggestions | None, short, common, missing class, and fallback | All five returns |

## Commands

```
coverage erase
coverage run --branch --source=password_policy -m pytest -q
BranchCoverageTest_39_password_policy.py
coverage report -m
```

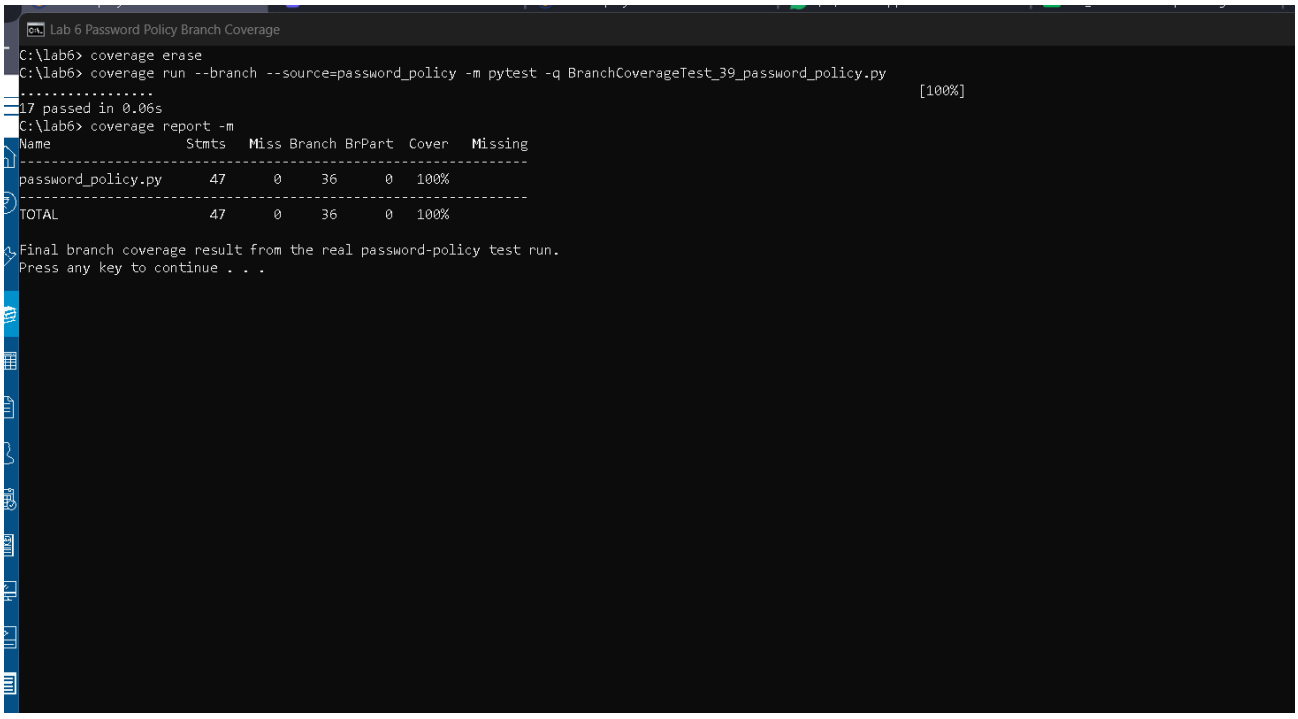



*Figure 2 Genuine terminal screenshot showing 17 passing tests and 100% branch coverage*

The final report records 47 statements with no misses and 36 branches with no partial branches. The result is 100% combined coverage, which exceeds the required branch-coverage threshold.

## Results and Interpretation

Coverage improved in two deliberate stages. The starter tests established that the main valid-password path worked, while the 91% suite covered most executable behavior. The final suite closed every remaining branch by targeting the specific outcomes reported as missing.

| Metric | Starter | Code milestone | Final branch milestone |
|---|---|---|---|
| Tests passed | 2 | 6 | 17 |
| Statements | 26 of 47 | 43 of 47 | 47 of 47 |
| Branches | 9 partial | Not measured | 36 of 36 |
| Coverage | 52% combined | 91% line | 100% combined |

## Repository and Commit Evidence

**Completed repository:** https://github.com/Adi-2936/lab-6-code-branch-coverage

| Commit | Purpose | Coverage evidence |
|---|---|---|
| 80a8724 | Replace sample with assigned password-policy code and code tests | 91% line |
| 86840d5 | Add complete branch tests and genuine terminal screenshot | 100% branch |

## Conclusion

The assigned password-policy module now satisfies every Lab 6 deliverable. The repository contains separate code-coverage and branch-coverage test files, genuine terminal screenshots, and distinct commits for the two milestones. Final branch analysis confirms that every statement and all 36 recorded branch outcomes execute successfully.